

YINUO HAN

1200 E. California Blvd., Pasadena, CA 91125, USA
yinuo@caltech.edu · yinuohan.github.io

RESEARCH INTERESTS

- Exoplanetary systems, debris disks, planet formation, disk substructure, dynamical evolution
- High-angular-resolution imaging, high-contrast optical/infrared imaging, millimeter interferometry
- Dust formation, carbon feedback, Wolf-Rayet stars, colliding-wind binaries

ACADEMIC APPOINTMENTS

1. **Melza M. and Frank Theodore Barr Fellow, Caltech, USA** Since Oct 2024
Postdoctoral fellowship, Planetary Sciences, Division of Geological and Planetary Sciences

EDUCATION

1. **PhD in Astronomy, University of Cambridge, UK** Oct 2020 - Jul 2024
Institute of Astronomy · Trinity College · Gates Cambridge Scholar
Thesis: *The Structure of Extrasolar Planetesimal Belts in Images*
Advisor: Prof. Mark Wyatt
2. **BSc (Adv) (Hons), University of Sydney, Australia** Feb 2016 - Dec 2019
Majors in Physics, Neuroscience · First Class Honours (Physics) and the University Medal
Thesis: *A Dance with Dragons: The Enigmatic Wolf-Rayet Binary Apep*
Advisor: Prof. Peter Tuthill
- Study Abroad, Physics, University of California, Berkeley, USA** Aug 2017 - Dec 2017

ADVISING

List of theses and research projects advised.

- *VLT observations of β Pic and its dust clump.* Jun – Sep 2025
Summer research project/Caltech SURF by Polaris Hayes (Caltech).
- *ALMA imaging of WR112 and dust modeling with JWST/MIRI.* Jun – Sep 2025
Summer research project/Caltech SURF by Donglin Wu (Yale University).
- *Joint ALMA and JWST/NIRCam modeling of the vertical structure of the AU Mic debris disk.* Jun – Sep 2025
Summer research project/Caltech SURF by Junyi Zhao (University of Michigan).
- *Secular perturbation as the origin of the dust and gas clump of Beta Pictoris.* Oct 2022 – Jun 2023
Cambridge Part III/Masters project by Faran Sarwar (co-advised with Mark Wyatt).
- *The structure of the Beta Pictoris debris disk inferred from multi-wavelength mid-infrared images.* Jul – Aug 2022
Summer research project by Andrew Zhang.

TEACHING

1. **Topics in Astrophysics** (Distributions, Timescales, Tides, Planet Formation) 2023
Institute of Astronomy, University of Cambridge
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
2. **Structure and Evolution of Stars** 2022
Institute of Astronomy, University of Cambridge
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
3. **Oscillating Systems, Waves and Quantum Waves, Fields** 2021
Trinity College, University of Cambridge
Weekly supervisions/tutorial classes for Part IA Physics/first-year undergraduate physics.
4. **Physics 1B** (Electromagnetism, Fluids, Quantum Mechanics) 2019
School of Physics, University of Sydney
Weekly tutorial classes for first-year undergraduate physics.

OBSERVING

- *Keck P.I. program*: Probing planet–disk interactions in asymmetric debris disks 2026B
- *ALMA co-I. program*: Testing a new picture of debris disk gas by observing CI towards HD 121617 (ID 2025.1.00273.S, P.I. Gianni Cataldi) C12, 2025
- *VLT/VISIR P.I. program*: Testing the giant impact scenario to explain the dust clump and cat’s tail in Beta Pictoris (114.27CQ) P114, 2024
- *ALMA P.I. program*: First resolved ALMA imaging of carbon-rich Wolf-Rayet binary dust structures (2024.1.00803.S) C11, 2024
- *ALMA co-I. program*: Resolving the extended structure associated with Apep’s enigmatic wind (2024.1.01244.S, P.I. Joseph Callingham) C10, 2024
- *JWST P.I. program*: What lies beyond the inner spiral of Apep? (GO 5842) C3, 2024
- *JWST P.I. program*: What causes warm dust interior to planetesimal belts? (GO 5709) C3, 2024
- *VLT/VISIR P.I. program*: Wind kinematics in the extreme colliding-wind binary Apep: the first Galactic long-duration gamma-ray burst progenitor? (113.26A7) P113, 2024
- *ALMA P.I. program*: First ALMA imaging of dust around Wolf-Rayet binaries (2023.1.00999.S) C10, 2023
- *ALMA co-I. program*: Vertical structure and planetary system dynamics (2023.1.00487.S and 2025.1.00362.S, P.I. Meredith Hughes) C10/C12, 2023/2025
- *JWST co-I. program*: Fingerprinting the history of episodic dust creation in Wolf-Rayet binaries (4093, P.I. Noel Richardson) C2, 2023
- *Australia Telescope Compact Array*: Executing observations of Wolf-Rayet binary Apep 2019

PRESS

- *NRAO press release based on Donglin Wu's summer research published in Wu et al. 2026.* Feb 2026
Yale press release.
- *Caltech press release based on Han et al. 2026.* Jan 2026
Other ARKS collaboration press releases from NRAO, ALMA and ERC.
- *NASA, STScI and Caltech press releases based on Han et al. 2025.* Nov 2025
Featured on Astronomy Picture of the Day.
- *University of Cambridge press release based on Han et al. 2022.* Oct 2022
Reported by major science and mainstream news outlets (e.g., ABC, BBC, CNN, The Guardian).
- *University of Sydney press release based on Han et al. 2020.* Oct 2020
Reported by 110 news items from 21 countries in 8 languages (e.g., CNN).

ACADEMIC AWARDS

UNIVERSITY OF CAMBRIDGE

- Gates Cambridge Scholarship* 2020
Cambridge International Scholarship, accepted Gates above instead 2020

UNIVERSITY OF SYDNEY

- University Medal*, top award among graduating class in physics 2019
Henry Chamberlain Russell Prize for Astronomy, for best astronomy Honours thesis 2019
University of Sydney Honours Scholarship 2019
Faculty of Science Olympiad Scholarship 2016 - 2019
Deas-Thomson Scholarship in Physics, for top of class in third-year physics 2018
W.I.B. Smith Prize in Experimental Physics 2018
Dean's List of Academic Excellence 2016, 2017, 2018
Sydney Scholars Award 2016 - 2018
Denison Summer Research Scholarship 2016, 2017
Sydney Abroad International Exchange Scholarship, supported second-year physics exchange 2017
University of Sydney Academic Merit Prize 2016
Levey Scholarship in Physics, for top of class in first-year physics 2016
Smith Prize in Experimental Physics 2016

OTHER

- University of Oxford Clarendon Scholarship*, declined 2020
International Physics Olympiad Bronze Medal 2015

RESEARCH GRANTS

EXTERNAL GRANTS

- P.I., JWST Program 5842 (USD 78k) 2024 – 2027
P.I., JWST Program 5709 (USD 74k) 2024 – 2027

INTERNAL GRANTS

Barr Fellowship research grant, Caltech (USD 30k)	2024 – 2027
Gates Cambridge Scholarship research funding, University of Cambridge (approx. USD 3k)	2020 – 2024
Rouse Ball research funding, Trinity College, Cambridge (approx. USD 4k)	2023

COLLABORATIONS

Major collaborations on large observational campaigns.

ARKS: ALMA Large Program to image debris disks at high resolution, P.I. Sebastian Marino

- *Role: Leading Radial Structure team. Our findings are available on arkslp.org.*

JWST High Contrast Imaging: JWST ERS program imaging exoplanets and disks, P.I. Sasha Hinkley

- *Role: Modeling MIRI 4-quadrant phase mask observations and VLT observations of HD 141569*

DustERS: JWST ERS program on Wolf-Rayet binary dust formation, P.I. Ryan Lau

- *Role: Developing spiral dust model for MIRI imaging published in Lau et al. 2022.*

TALKS

CONFERENCE TALKS

- Exoplanets and Planet Formation Dec 2025
The high-resolution structure of debris disks in the ARKS ALMA program. Shanghai, China
- Exploring Star–Planet–Disk Connections Oct 2025
High-resolution ALMA imaging of extrasolar planetesimal belts. Tartu, Estonia
- Europlanet Science Congress Sep 2024
The high-resolution structure of debris disks revealed by the ARKS ALMA program. Berlin, Germany
- Raising the veil on star formation near and far Apr 2024
High-resolution ALMA imaging of debris disks with the ARKS program. Cambridge, UK
- Dust Devils: Debris Disks in the Sonoran Desert Mar 2024
The radial structure of debris disks in the ARKS ALMA program. Tucson, USA
- Cosmic Dust Aug 2023
The physics of dusty Wolf-Rayet binaries revealed by observations of WR140. Kitakyushu, Japan
- UK Exoplanet Community Meeting Sep 2022
Recovering the structure of edge-on debris disks non-parametrically. Edinburgh, UK
- Debris Disks at Home and Abroad Sep 2022
Recovering the structure of edge-on debris disks non-parametrically. Jena, Germany

SEMINARS

- Macquarie University May
Observing extrasolar comets Sydney, Australia
- Institute for Astronomy, University of Hawaii Apr 2026
Understanding planetary system formation and evolution through the lens of debris disks Manoa HI, USA
- Caltech DIX Planetary Science Seminar Mar 2026
Imaging Extrasolar Kuiper Belts and Interpreting Their Structures Pasadena CA, USA

- MPIA Planet and Star Formation Seminar
The formation and evolution of dust in the colliding-wind binary Apep revealed by JWST. Oct 2025
Heidelberg, Germany
- Tsinghua Exoplanets and Planet Formation Seminar
Imaging debris disks with ALMA and JWST. Sep 2025
Beijing, China
- Columbia Astronomy & Astrophysics Seminar
Resolving radial substructures in debris disks. Mar 2025
New York City, USA
- JPL Astrophysics Luncheon Seminar
Interpreting the Resolved Structure of Debris Disks. Feb 2025
Pasadena, USA
- Department of Applied Mathematics and Theoretical Physics, University of Cambridge
The radial and vertical structure of debris disks. Oct 2023
Cambridge, UK
- Institute of Astronomy, University of Cambridge
Stellar fireworks: the spiral dust plumes of Wolf-Rayet binaries. Oct 2022
Cambridge, UK
- Institute of Astronomy, University of Cambridge
A non-parametric method to recover the structure of edge-on debris disks. Mar 2022
Cambridge, UK

OUTREACH TALKS

- Institute of Astronomy Open Evening
What are exoplanets doing? Asteroid belts give us a clue. Oct 2023
Cambridge, UK
- Institute of Astronomy Open Evening
Stellar fireworks: a journey through the spirally dust of massive binary stars. Jan 2023
Cambridge, UK
- Integrated Science Class Seminar (Year 6–8), Ming Cheng School
Planets of the Solar System and beyond. Apr 2022
Beijing, China
- Gates Cambridge Teach-a-thon
Planets of the Solar System and beyond. Feb 2021
Online
- Australian Council of Undergraduate Research
Highly anisotropic stellar winds in recently-identified Wolf-Rayet binary. Oct 2019
Newcastle, Australia

POSTER PRESENTATIONS

- Gordon Research Conference on the Origins of Solar Systems
The radial structure of debris disks in the ARKS ALMA program. Jun 2025
South Hadley, USA
- Gordon Research Seminar + Conference on the Origins of Solar Systems
Modeling the structure of debris disks and the dust clump of Beta Pictoris. Jun 2023
South Hadley, USA
- STScI Spring Symposium: Planetary Systems and the Origins of Life in the Era of JWST
Modeling the structure of debris disks and the dust clump of Beta Pictoris. May 2023
Baltimore, USA
- Protostars and Planets VII
Has the dust clump in the debris disk of Beta Pictoris moved? Apr 2023
Kyoto, Japan

PUBLICATIONS

Publications available on ORCID 0000-0002-2106-0403, ADS Library and Google Scholar.

FIRST-AUTHOR PUBLICATIONS

9. **Y. Han**, M. C. Wyatt, K. Y. L. Su, *et al.* A radially broad collisional cascade in the debris disk of γ Oph revealed by JWST. *The Astrophysical Journal* (2026). doi.org/10.3847/1538-4357/ae6ef7
8. **Y. Han**, M. C. Wyatt, M. Jankovic, *et al.* The multiwavelength vertical structure of the archetypal β Pictoris debris disk. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202659709
7. **Y. Han**, E. Mansell, J. Jennings, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) II. The Radial Structure of Debris Disks. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556450
6. **Y. Han**, R. M. T. White, J. R. Callingham, *et al.* The formation and evolution of dust in the colliding-wind Wolf-Rayet binary Apep observed by JWST. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/ae12e5
5. **Y. Han**, M. C. Wyatt, S. Marino. Recovering the structure of debris disks non-parametrically from images. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf282
4. **Y. Han**, M. C. Wyatt, W. R. F. Dent. Has the dust clump in the debris disk of Beta Pictoris moved? *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stac3769
3. **Y. Han**, P. G. Tuthill, R. M. Lau, A. Soulain. Radiation-driven acceleration in the expanding WR140 dust shell. *Nature* (2022). doi.org/10.1038/s41586-022-05155-5
2. **Y. Han**, M. C. Wyatt, L. Matr . RAVE: a non-parametric method for recovering the surface brightness and height profiles of edge-on debris disks. *Monthly Notices of the Royal Astronomical Society* (2022). doi.org/10.1093/mnras/stac373
1. **Y. Han**, P. G. Tuthill, A. Soulain, J. R. Callingham, P. M. Williams, P. A. Crowther, B. J. S. Pope, B. Marcote. The extreme colliding-wind system Apep: resolved imagery of the central binary and dust plume in the infrared. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa2349

CO-AUTHORED PUBLICATIONS

* *Indicates projects led by students who I advised*

26. [In review] E. Chiang, T. D. Pearce, M. R. Jankovic, A. J. Backues, **Y. Han**, A. V. Krivov, M. Pan, B. Zawadzki, *et al.* Viscously Stirring Particle Disks into Lorentzians and Gaussians to Infer Dynamical and Collisional Masses (ARKS XIII). *The Astrophysical Journal* (2026). arxiv.org/abs/2604.19743
25. J. Olofsson, M. R. Jankovic, S. Marino, A. V. Krivov, M. Bonduelle, G. Cataldi, **Y. Han**, A. M. Hughes, T. L hne, S. Mac Manamon, E. Mansell, L. Matr , J. Milli, A. A. Sefilian, P. Th bault, B. Zawadzki, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) XI: Gas-dust interactions and radial offsets between micron and millimetre-sized grains. *Astronomy & Astrophysics* (2026). arxiv.org/abs/2605.03009

24. * D. Wu, **Y. Han**, *et al.* Constraining properties of dust formed in Wolf-Rayet binary WR 112 using mid-infrared and millimeter observations. *The Astrophysical Journal* (2026). doi.org/10.3847/1538-4357/ae31f1
23. S. Marino, L. Matr , A. M. Hughes, J. Ehrhardt, G. M. Kennedy, C. del Burgo, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) I. Motivation, Sample, Data Reduction and Results Overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556489
22. B. Zawadzki, A. Fehr, A. M. Hughes, E. Mansell, J. Kittling, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) III. The Vertical Structure of Debris Discs. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556505
21. S. Mac Manamon, L. Matr , S. Marino, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IV. CO gas imaging and overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556605
20. J. Milli, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) V. Comparison between scattered light and thermal emission. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556523
19. J. B. Lovell, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VI. Asymmetries and Offsets. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556568
18. A. Brennan, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VII. Optically Thick Gas with Broad CO Gaussian Local Line Profiles in the HD 121617 Disc. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556566
17. S. Marino, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VIII. A dust arc and non-Keplerian gas kinematics in HD 121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556493
16. P. Weber, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IX. Gas-driven origin for the continuum arc in the debris disc of HD121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556855
15. M. Jankovic, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) X. Interpreting the peculiar dust rings around HD 131835. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556637
14. R. M. T. White, B. J. S. Pope, P. G. Tuthill, **Y. Han**, *et al.* The Serpent Eating Its Own Tail: Dust Destruction in the Apep Colliding-Wind Nebula. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/adfbel
13. J. D. Monnier, **Y. Han**, *et al.* Revealing the Accelerating Wind in the Inner Region of Colliding-wind Binary WR 112. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-3881/adfa03
12. N. D. Richardson, *et al.* Carbon-rich dust injected into the interstellar medium by Galactic WC binaries survives for hundreds of years. *The Astrophysical Journal* (2025). 10.3847/1538-4357/addf30

11. M. Sommer, M. C. Wyatt, **Y. Han**. A PR drag origin for the Fomalhaut disk’s pervasive inner dust: constraints on collisional strengths, icy composition, and embedded planets. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf494
10. E. P. Lieb, *et al.* Dynamic imprints of colliding-wind dust formation from WR140. *The Astrophysical Journal Letters* (2025). doi.org/10.3847/2041-8213/ad9aa9
9. R. M. Lau, *et al.* A first look with JWST aperture masking interferometry (AMI): resolving circumstellar dust around the Wolf-Rayet binary WR 137 beyond the Rayleigh limit. *The Astrophysical Journal* (2024). doi.org/10.3847/2041-8213/ad9aa9
8. J. Terrill, S. Marino, R. A. Booth, **Y. Han**, J. Jennings, M. C. Wyatt. Deprojecting and constraining the vertical thickness of exoKuiper belts. *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stad1847
7. R. M. Lau, *et al.* From dust to nanodust: resolving circumstellar dust from the colliding-wind binary Wolf-Rayet 140. *The Astrophysical Journal* (2022). doi.org/10.3847/1538-4357/acd4c5
6. R. M. Lau, M. J. Hankins, **Y. Han**, *et al.* Nested dust shells around the Wolf-Rayet binary WR 140 observed with JWST. *Nature Astronomy* (2022). doi.org/10.1038/s41550-022-01812-x
5. P. A. Robinson, X. Gao, **Y. Han**. Relationships between lognormal distributions of neural properties, activity, criticality, and connectivity. *Biological Cybernetics* (2021). doi.org/10.1007/s00422-021-00871-z
4. B. Marcote, J. R. Callingham, M. De Becker, P. G. Edwards, **Y. Han**, R. Schulz, J. Stevens, P. G. Tuthill. AU-scale radio imaging of the wind collision region in the brightest and most luminous non-thermal colliding wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2021). doi.org/10.1093/mnras/staa3863
3. R. M. Lau, M. J. Hankins, **Y. Han** *et al.* Resolving periodic spirals and shadows from the Wolf-Rayet dust factory WR112. *The Astrophysical Journal* (2020). doi.org/10.3847/1538-4357/abaab8
2. J. R. Callingham, P. A. Crowther, P. M. Williams, P. G. Tuthill, **Y. Han**, B. J. S. Pope. Two Wolf-Rayet stars at the heart of colliding-wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa1244
1. M. Li, **Y. Han**, M. J. Aburn, M. Breakspear, R. Q. Poldrack, J. M. Shine, J. T. Lizier. Transitions in information processing dynamics at the whole-brain network level are driven by alterations in neural gain. *PLOS Computational Biology* (2019). doi.org/10.1371/journal.pcbi.1006957